

Q. Chapter-02: Boolean Algebra & Logic Gate.

* Ex-2.1: Which of the six basic laws (closure, Associative, commutative, Identity, Inverse and distributive) are satisfied for the pair of binary operators listed below?

+	0	1	2
0	0	0	0
1	0	1	1
2	0	1	2

	0	1	2
0	0	1	2
1	1	1	2
2	2	2	2

Solⁿ:

$\Rightarrow$ closure, $\Rightarrow$ Associative $\Rightarrow$ commutative
 $\Rightarrow$ distributive $\Rightarrow$ Identity for + is 2
 $\Rightarrow$ Identity for $\cdot$ is 0 $\Rightarrow$ no inverse.

* Ex-2.2: Show that, the set of three elements $\{0, 1, 2\}$ and the two binary operators + and $\cdot$ as defined by the above table is ~~not~~ not a Boolean algebra.

State which of the Huntington ⁵⁰ postulates is not satisfied?

Ans: All postulates are satisfied except for postulate 5; there is no complement.

* Ex-2.3: Demonstrate by means of truth tables the validity of following theorems of Boolean algebra.

(a) The associative laws: The associative laws of Boolean algebra are:

$$(A + B) + C = A + (B + C)$$

$$(A \times B) \times C = A \times (B \times C)$$

⇒ Truth table for

$$(A + B) + C = A + (B + C)$$

A	B	C	$A+B$	$(A+B)+C$	$B+C$	$A+(B+C)$
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	1	1	0	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

from above truth table we find
 $(A+B)+C = A+(B+C)$ (proved).

⇒ Truth table for $(A \times B) \times C = A \times (B \times C)$

A	B	C	$A \times B$	$(A \times B) \times C$	$B \times C$	$A \times (B \times C)$
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	1	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	1	0	0	0
1	1	1	1	1	1	1

$$(A \times B) \times C = A \times (B \times C)$$

(b) De Morgan's theorems for three 52

variables:

$$\overline{(A+B+C)} = \bar{A} \times \bar{B} \times \bar{C}$$

$$\overline{(A \times B \times C)} = \bar{A} + \bar{B} + \bar{C}$$

⇒ Truth table for $\overline{(A+B+C)} = \bar{A} \times \bar{B} \times \bar{C}$

A	B	C	A+B+C	$\overline{(A+B+C)}$	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{A} \times \bar{B} \times \bar{C}$
0	0	0	0	1	1	1	1	1
0	0	1	1	0	1	1	0	0
0	1	0	1	0	1	0	1	0
0	1	1	1	0	1	0	0	0
1	0	0	1	0	0	1	1	0
1	0	1	1	0	0	1	0	0
1	1	0	1	0	0	0	1	0
1	1	1	1	0	0	0	0	0

$$\overline{(A \times B \times C)} = \bar{A} + \bar{B} + \bar{C}$$

⇒ Truth table for $\overline{(A \times B \times C)} = \bar{A} + \bar{B} + \bar{C}$

A	B	C	A×B×C	$\overline{(A \times B \times C)}$	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{A} + \bar{B} + \bar{C}$
0	0	0	0	1	1	1	1	1
0	0	1	0	1	1	0	1	1
0	1	0	0	1	1	0	1	1
0	1	1	0	1	1	0	0	1
1	0	0	0	1	0	1	1	1
1	0	1	0	1	0	1	0	1
1	1	0	0	1	0	0	1	1
1	1	1	1	0	0	0	0	0

A	B	C	$A \times B \times C$	$\overline{A \times B \times C}$	A	B	C	$A + B + C$
1	1	0	0	1	0	0	1	1
1	1	1	1	0	0	0	0	0

(C) The distributive law of $+$ over:

$$A + (BC) = (A + B)(A + C)$$

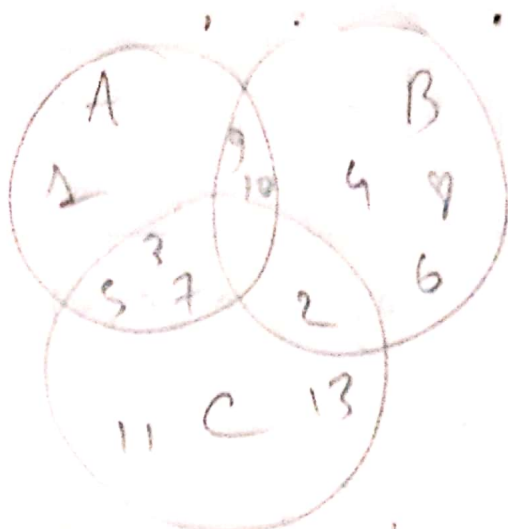
A	B	C	BC	$A + (BC)$	$A + B$	$A + C$	$(A + B)(A + C)$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0
0	1	0	0	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	1	1	1	1	1

*Ex-2.4: Repeat problem 2.3 using Venn diagrams.

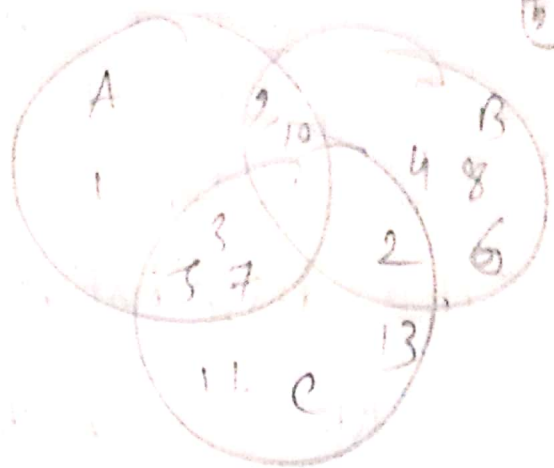
(A) Associative law:

$$A \cup (B \cap C) = (A \cup B) \cap C$$

$$A \cap (B \cup C) = (A \cap B) \cup C$$



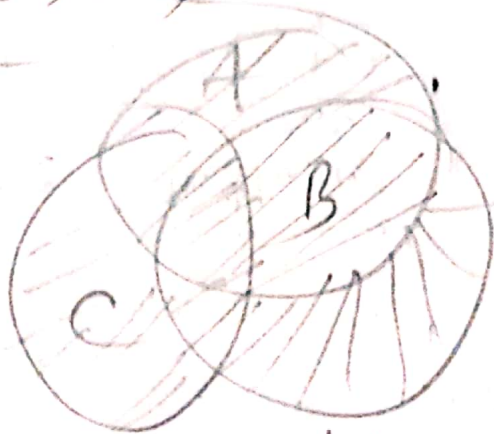
$A \cup B \cup C$



$(A \cup B) \cup C$

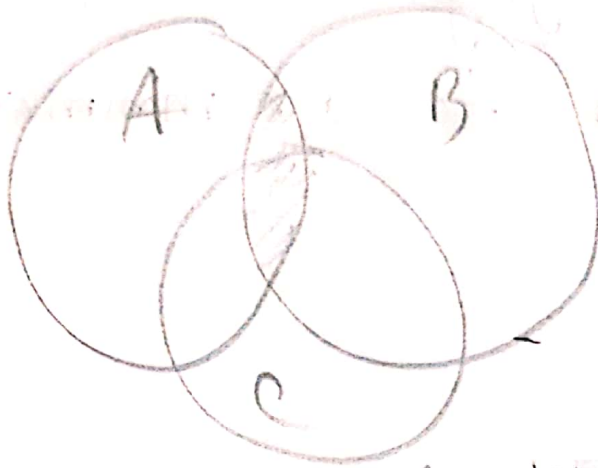
(b) De Morgan's theorem for three variables:

$$\Rightarrow (A \cup B \cup C)' = A' \cap B' \cap C'$$



$$(A \cup B \cup C)' = A' \cap B' \cap C'$$

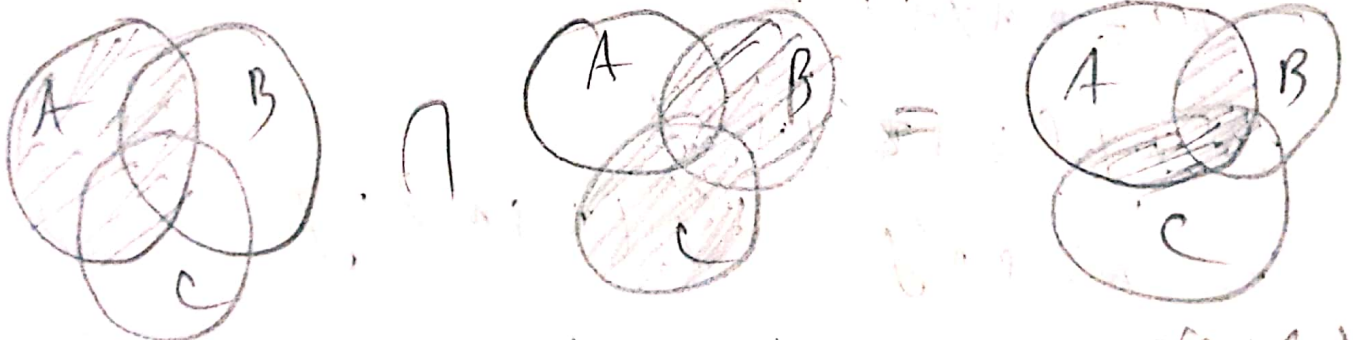
$$\Rightarrow (A \cap B \cap C)' = A' \cup B' \cup C'$$



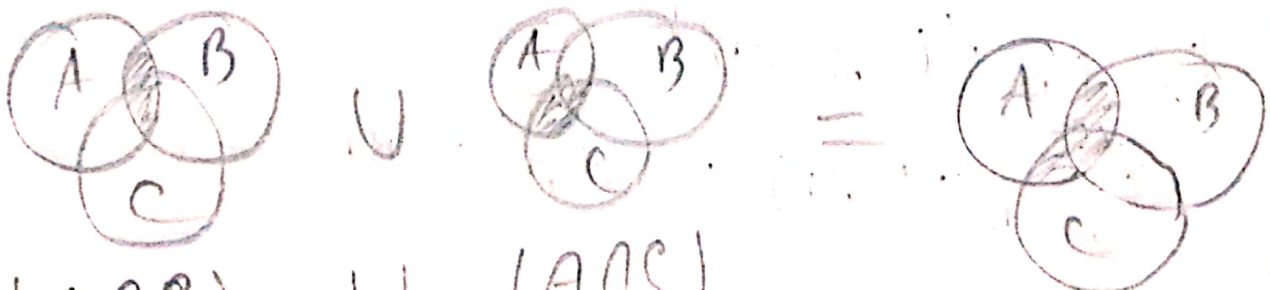
$$(A \cap B \cap C)' = A' \cup B' \cup C'$$

(C) Distributive law:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$



$$(A) \cap (B \cup C) = A \cap (B \cup C)$$



$$(A \cap B) \cup (A \cap C) = (A \cap (B \cup C))$$

* Ex-2.5: Simplifying the following Boolean functions to a minimum number of literals.

$$(a) \quad xy + xy' \\ = x(y + y') = x$$

$$(b) \quad (x + y)(x + y') \\ = xx + xy' + xy + yy' \\ = x + xy' + xy = x + x(y + y') = x + x = x$$

$$(c) \quad xyz + x'y + xyz' \\ = xy(z + z') + x'y \\ = xy + x'y = y(x + x') = y$$

$$(d) \quad zx + zx' = z(x + x') = z(x + y)$$

$$(e) \quad (A + B)'(A' + B')' \\ = (A'B')(AB) = AA' \times BB' = 0$$

$$(f) \quad y(wz' + wz) + xy \\ = yw(z + z') + xy = yw + xy = y(x + w)$$

* Ex-2.6: Reduce the following Boolean expressions to the required number of literals.

(a) $ABC + A'B'C + A'BC + ABC' + A'B'C'$ to five number of literals.

⇒ we used K-map for simplify the expression.

BC \ A	00	01	11	10
0	1	1	1	
1			1	1

$$F_1 = A'B', F_2 = BC, F_3 = AB$$

$$\therefore F = AB + A'B' + BC = A'B' + B(A+C)$$

(b) $BC + AC' + AB + BCD$ to four literals

Soln: we use K-map same as problem (a) and we find

$$= AC' + BC$$

Answer

(c) $[(cD)' + A]' + A + cD + AB$ to three literals.

Solⁿ: we used k-map and find the simplified form.

$$F = A + cD$$

(d) $(A + c + D)(A + c + D')(A + c + D)(A + B)'$ to four literals.

Solⁿ: we used k-map same as (a) and find,

$$F = A + BCD.$$

* Ex-2.7: Find the complement of the following Boolean functions and reduce them to a minimum number of literals.

$$(a) (BC' + A'D) (AB + CD')$$

(59)

$$= AB'BC' + AA'B'D + CC'D'B + DD'A'C$$

$$= 0 + 0 + 0 + 0 \quad [\because xx = 0]$$

$$= 0$$

$$(b) B'D + A'BC' + ACD + A'BC$$

$$= A'BC' + A'BC + B'D + ACD$$

$$= A'B(C + C') + B'D + ACD$$

$$= A'B + B'D + ACD$$

$$(c) [(AB)'A] [(AB)'B]$$

$$= [(A' + B')A] [(A' + B')B]$$

$$= (A'A + AB') (A'B + BB')$$

$$= (0 + AB') (A'B + 0)$$

$$= AB' \times A'B$$

$$= AA' \times BB'$$

$$= 0$$

$$(d) \quad AB' + C'D'$$

$$= (A' + B)(C + D)$$

(10)

*Ex - 2.8: Given two Boolean functions F_1 and F_2 :

(a) Show that the boolean function $E = F_1 + F_2$ obtained by OR'ing the two functions, contains the sum of all the minterms in F_1 and F_2 .

Solⁿ: $F_1 = \sum m_i$ and $F_2 = \sum m_j$

$$E = F_1 + F_2$$

$$= \sum m_i + m_j = \sum (m_i + m_j)$$

(b) Show that the boolean function $G = F_1 F_2$, obtained from AND'ing the two functions, contains those min-terms common to both F_1 and F_2 .

Solⁿ:

$$F_1 = \sum m_i, F_2 = \sum m_j$$

(61)

$$\therefore G = F_1 \cdot F_2$$

$$= \sum m_i \cdot \sum m_j$$

$$m_i \cdot m_j = 0, \text{ if } i \neq j$$

$$m_i \cdot m_j = 1, \text{ if } i = j$$

Therefore, G has only common minterms.

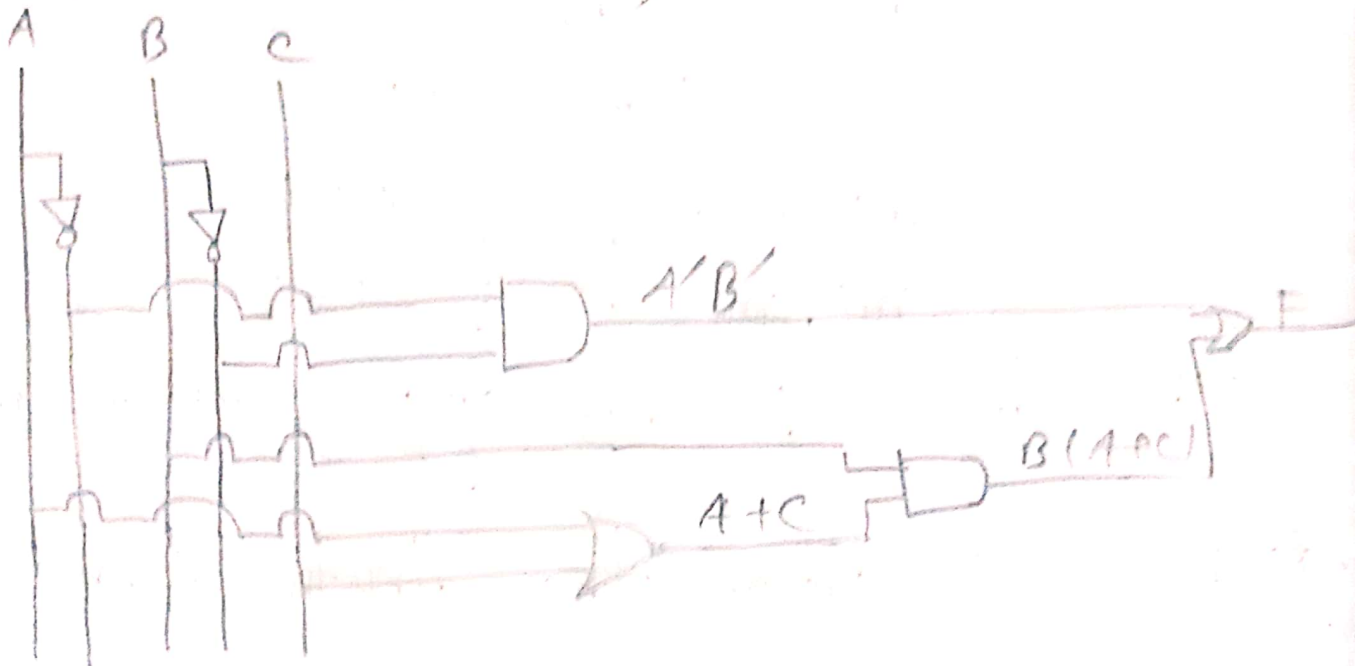
* Ex-2.9: obtain the truth table of the functions: $F = xy + x'y' + y'z$

x	y	z	xy	x'y'	y'z	$F = xy + x'y' + y'z$
0	0	0	0	0	0	0
0	0	1	0	0	1	1
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	0	1	0	1
1	0	1	0	1	1	1
1	1	0	1	0	0	1
1	1	1	1	0	0	1

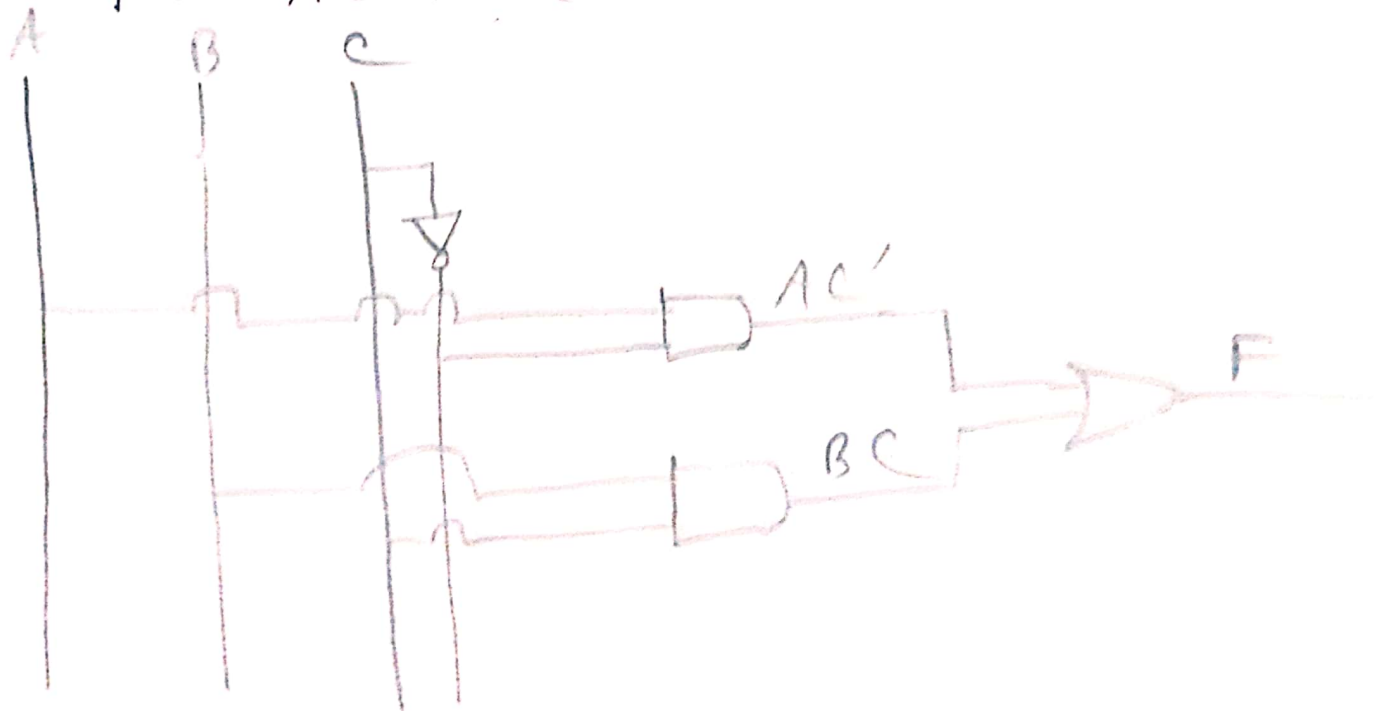
* Ex-2.10: Implement the simplified Boolean functions from problem 2.6 with logic gates

Ex Sol^m:

(a) $F = A'B' + B(A+C)$

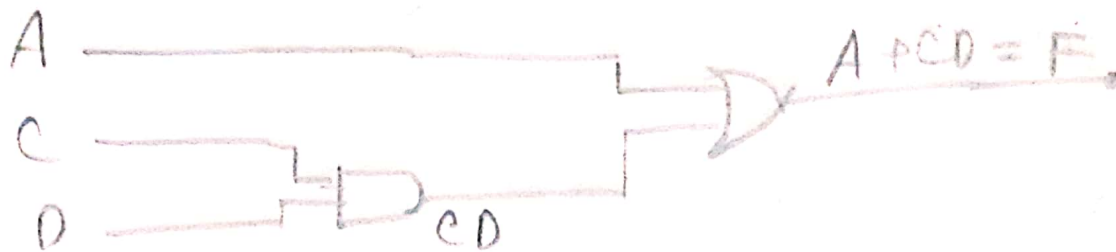


(b) $F = AC' + BC$

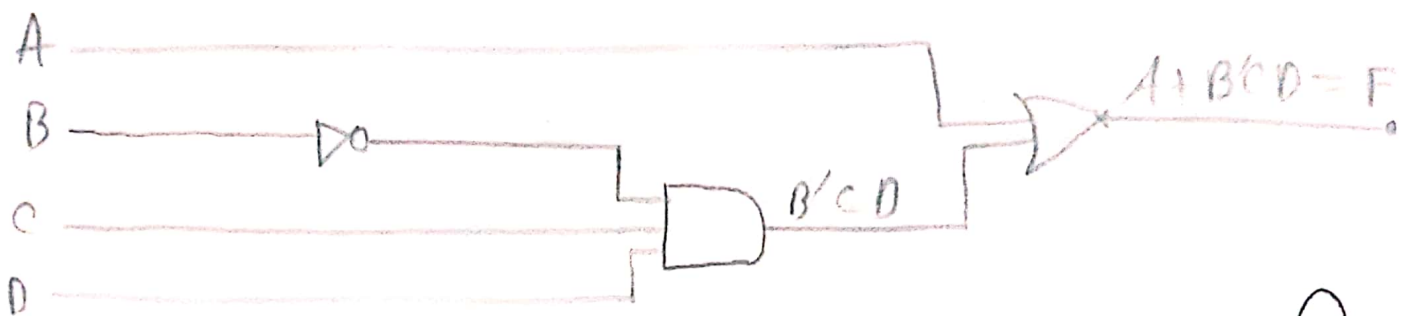


(c) $F = A + CD$

(63)



(d) $F = A + B'CD$

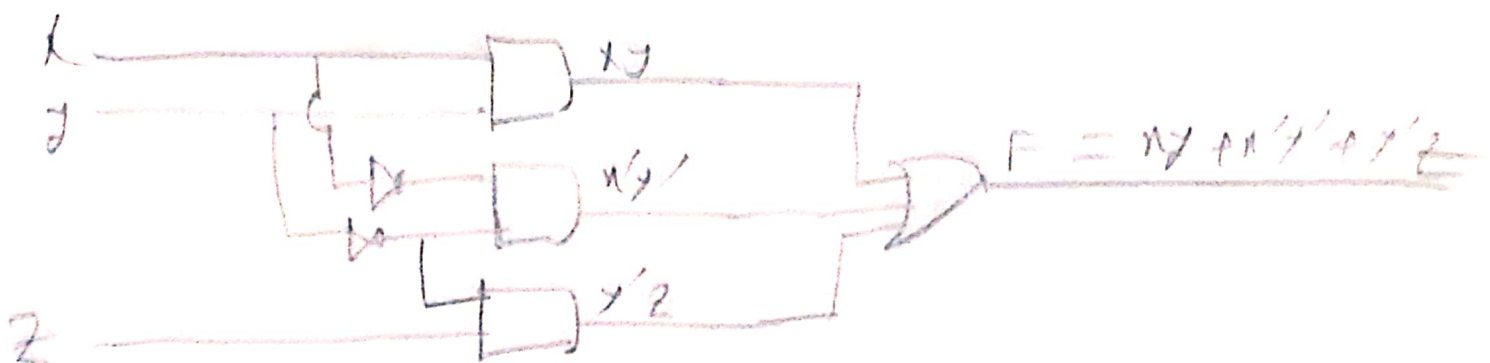


* E.X-2.11: Given the boolean function

$$F = xy + x'y' + y'z$$

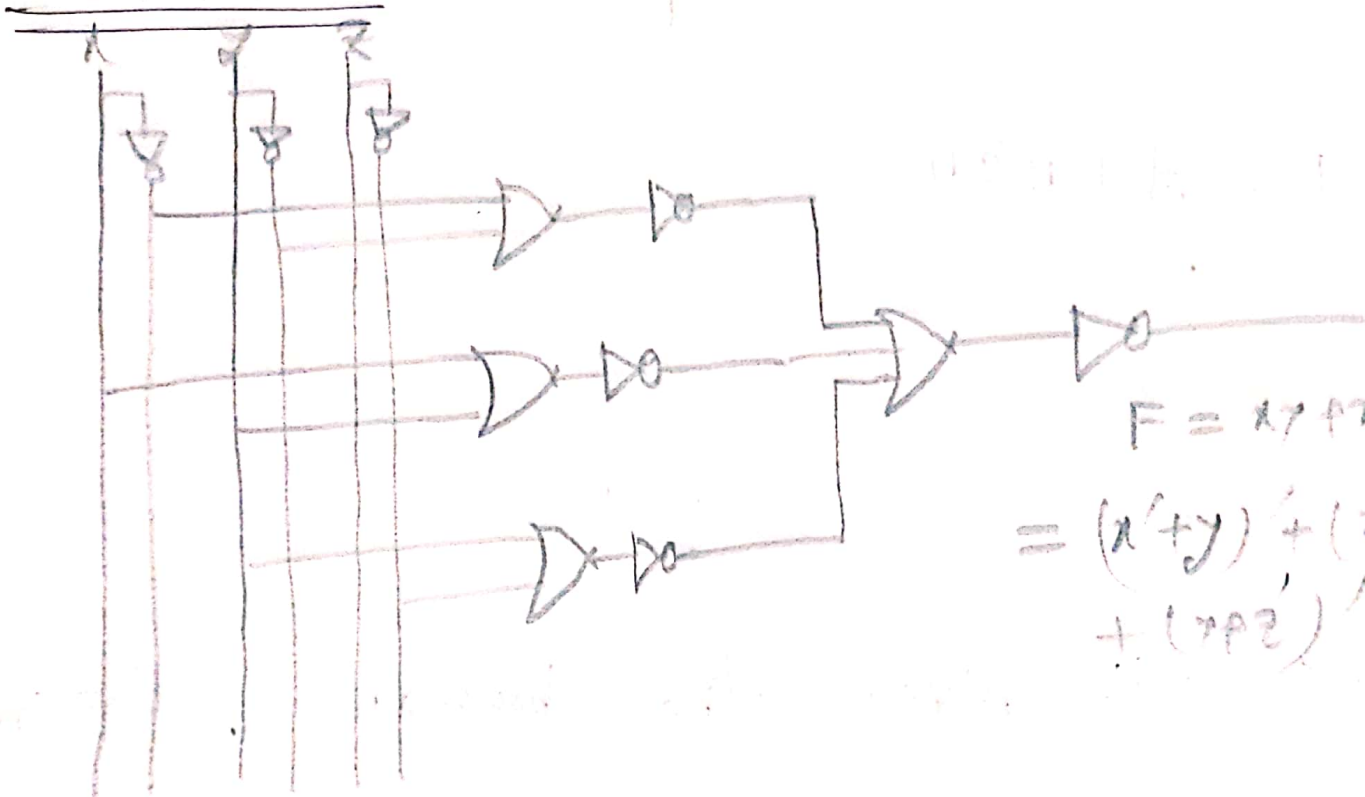
(a) Implement it with AND, OR and NOT Gates,

Solⁿ:



(b) Implement it with only OR and NOT Gate:

Solⁿ:



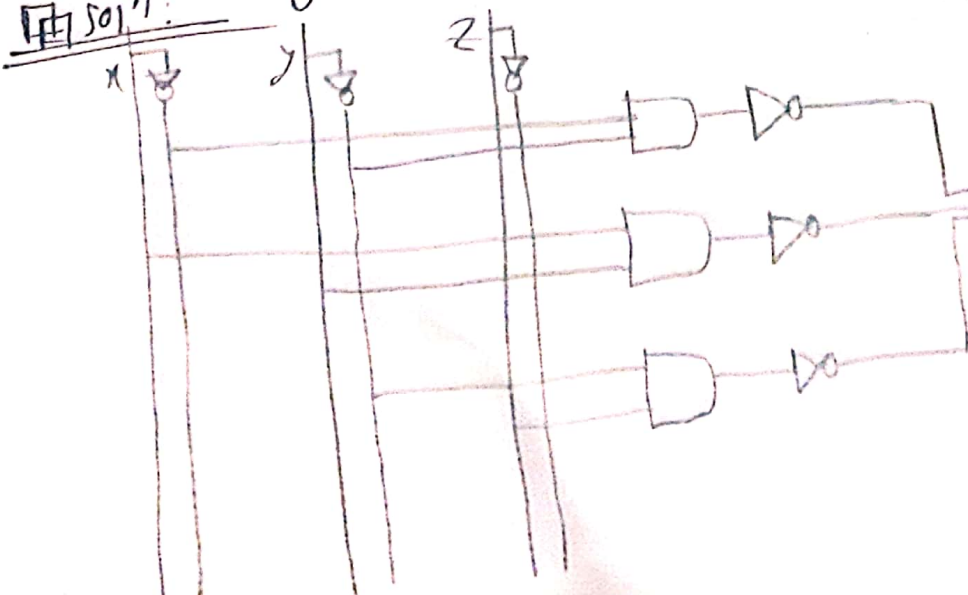
$$F = xy + x'y' + y'z$$

$$= (x' + y)' + (x + y')'$$

$$+ (x + yz)'$$

(c) Implement it with only AND and NOT gates.

Solⁿ:



$$F = xy + x'y' + y'z$$

$$= [(x + y)(x' + y')(x' + z)]'$$

* Ex-2.12: Simplify the functions T_1 and T_2 to a minimum number of literals.

A	B	C	T_1	T_2
0	0	0	1	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

Solⁿ:

$$\begin{aligned}
 T_1 &= A'B'C' + A'B'C + A'BC' \\
 &= A'B'(C + C') + A'BC' \\
 &= A'B' + A'BC' \\
 &= (A'B' + A'B)(A'B' + C') \\
 &= A'(B + B')(A'B' + C') \\
 &= A'(A'B' + C') \\
 &= A'A'B' + A'C' \\
 &= A'B' + A'C' = A'(B' + C')
 \end{aligned}$$

Answer.

for T_2 :

$$\begin{aligned} T_2 &= (A+B+C) (A+B+C') (A+B'+C) \\ &= (A+AB+AC'+AB+BB+BC'+AC+BC+CC') (A+B'+C) \\ &= A(1+B+C'+C) + B(1+C'+C) + CC' \cdot 0 \\ &= (A+B) (A+B'+C) \\ &= A+AB'+AC+AB+BB' \\ &= A+BC \end{aligned}$$

Answer.

* Ex-2.13: Express the following function in a sum of minterms and a product of max-terms.

$$(a) F(A, B, C, D) = D(A'+B) + B'D$$

Ex Solⁿ: $F(A, B, C, D) = B'D + D(A' + B)$ (67)

$$= B'D + A'D + BD$$

$$\therefore \bar{B}D = (0+1)(0)(0+1)(1)$$

$$= 0 \ 0 \ 0 \ 1 (\bar{A} \bar{B} \bar{C} D) \Rightarrow 1$$

$$= 0 \ 0 \ 1 \ 1 (\bar{A} \bar{B} C D) \Rightarrow 3$$

$$= 1 \ 0 \ 0 \ 1 (A \bar{B} \bar{C} D) \Rightarrow 9$$

$$= 1 \ 0 \ 1 \ 1 (A \bar{B} C D) \Rightarrow 11$$

$$\bar{A}D = (0)(0+1)(0+1)(1)$$

$$= \bar{A} \bar{B} \bar{C} D$$

$$= 0 \ 0 \ 0 \ 1 (\bar{A} \bar{B} \bar{C} D) \Rightarrow 1$$

$$= 0 \ 0 \ 1 \ 1 (\bar{A} \bar{B} C D) \Rightarrow 3$$

$$= 0 \ 1 \ 0 \ 1 (\bar{A} B \bar{C} D) \Rightarrow 5$$

$$= 0 \ 1 \ 1 \ 1 (\bar{A} B C D) \Rightarrow 7$$

$$BD = (0+1)(1)(0+1)(1)$$

$$= 0 \ 1 \ 0 \ 1 (\bar{A} B \bar{C} D) \Rightarrow 5$$

$$= 0 \ 1 \ 1 \ 1 (\bar{A} B C D) \Rightarrow 7$$

$$= 1 \ 1 \ 0 \ 1 (A B \bar{C} D) \Rightarrow 13$$

$$= 1 \ 1 \ 1 \ 1 (A B C D) \Rightarrow 15$$

$$F(A, B, C, D) = \sum (1, 3, 5, 7, 9, 11, 13, 15)$$

$$= \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + \bar{A}BCD + A\bar{B}\bar{C}D + A\bar{B}CD + AB\bar{C}D + ABCD$$

⇒ product of maxterm (POS).

we know that complement of SOP is POS.

$$F(A, B, C, D) = \prod (0, 2, 4, 6, 8, 10, 12, 14)$$

$$= (A+B+C+D) \cdot (A+B+\bar{C}+D) \cdot (A+\bar{B}+C+D) \cdot (A+\bar{B}+\bar{C}+D) \cdot (\bar{A}+B+C+D) \cdot (\bar{A}+B+\bar{C}+D) \cdot (\bar{A}+\bar{B}+C+D) \cdot (\bar{A}+\bar{B}+\bar{C}+D)$$

$$(b) F(w, x, y, z) = y'z + wx y' + wx z' + w'x'z$$

So |ⁿ:

⇒ Sum of products (SOP):

$$F(w, x, y, z) = \sum (1, 3, 5, 9, 12, 13, 14)$$

⇒ product of sum (POS): we know that complement SOP is POS. (69)

$$POS = \prod (0, 2, 4, 6, 8, 10, 12, 15)$$

$$(C) \quad F(A, B, C, D) = (A + B' + C)(A + B')(A + C + D') \\ (A' + B + C + D)(B + C' + D')$$

⇒ Solⁿ: we can calculate it using solution (a) concept.

⇒ Sum of products (SOP):

$$F(A, B, C, D) = \sum (0, 1, 2, 8, 10, 12, 13, 14, 15)$$

⇒ products of sum (POS): we know that, complements of 'SOP' is 'POS'

$$F(A, B, C, D) = \prod (3, 4, 5, 6, 7, 9, 11)$$

$$(d) \quad F(A, B, C) = (A' + B)(B' + C)$$

⇒ Solⁿ:

⇒ Sum of products (SOP):

$$F(A, B, C) = \sum (0, 1, 3, 7)$$

$\Rightarrow$ product of sum (POS): we know that complement of SOP is POS.

$$F(A, B, C) = \prod(2, 4, 6).$$

(e) $F(x, y, z) = 1$

Solⁿ:

$\Rightarrow$ Sum of products (SOP):

$$F(x, y, z) = \sum(0, 1, 2, 3, 4, 6, 7)$$

$\Rightarrow$ product of sum (POS): There is no POS.

(f) $F(x, y, z) = (xy + z)(y + xz)$

Solⁿ:

$\Rightarrow$ Sum of products (SOP):

$$F(x, y, z) = \sum(3, 5, 6, 7)$$

$\Rightarrow$ product of sum (POS):

$$F(x, y, z) = \prod(0, 1, 2, 4),$$

* Ex-2.14: Convert the following to the other canonical form. (7)

(a) $F(x, y, z) = \Sigma(1, 3, 7)$

Soln:

$\Rightarrow$ canonical sum of product:

$$F(x, y, z) = x'y'z + x'yz + xyz$$

we can ^{write} like this,

$$F(x, y, z) = \Pi(0, 2, 4, 5, 6)$$

Canonical product of sum:

$$F(x, y, z) = (x + y + z) \cdot (x + y' + z) (x' + y + z) \\ (x' + y + z') (x' + y' + z)$$

(b) $F(A, B, C, D) = \Sigma(0, 2, 6, 11, 13, 14)$

Soln:

$\Rightarrow$ canonical sum of product:

$$F(A, B, C, D) = \Sigma(0, 2, 6, 11, 13, 14)$$

we can write like this,

$$F(A, B, C, D) = \Pi(1, 3, 4, 5, 7, 8, 9, 10, 12, 15)$$

$$(c) F(x, y, z) = \prod(0, 3, 6, 7)$$

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Solⁿ: Followed by a we find,

$$F(x, y, z) = \prod(0, 3, 6, 7).$$

we can write like this,

$$F(x, y, z) = \Sigma(1, 2, 4, 5)$$

$$(d) F(A, B, C, D) = \prod(0, 1, 2, 3, 4, 6, 12)$$

Solⁿ:

Canonical product of sum:

$$\Rightarrow F(A, B, C, D) = (A+B+C+D) \cdot (A+B+C+\bar{D}) \cdot (A+B+\bar{C}+D) \cdot (A+B+\bar{C}+\bar{D}) \cdot (A+\bar{B}+C+D) \cdot (A+\bar{B}+C+\bar{D}) \cdot (\bar{A}+B+C+D)$$

we can write this like,

$$F(A, B, C, D) = \Sigma(5, 7, 8, 9; 10, 11, 13, 14, 15)$$

Canonical sum of product,

$$F(A, B, C, D) = \bar{A}B\bar{C}D + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD + AB\bar{C}\bar{D} + AB\bar{C}D + ABC\bar{D} + ABCD$$

Answer:

* E.x - 2.15: what is the difference between canonical form and standard form? which form is preferable when implementing a Boolean function with gates? which form is obtained when reading a function from a truth table?

Solⁿ:

⇒ The main difference between canonical and standard form is that canonical form is a way of representing Boolean outputs of digital circuits using Boolean Algebra while standard form is a simplified version of canonical form that represents Boolean outputs of digital circuits using Boolean Algebra.

⇒ Standard form would be preferred over canonical form since it is more simplified and requires less gates.

⇒ Both canonical and standard form can be used to represent a Boolean function, but the standard form is generally preferred because it typically results in a simpler and more compact expression.

* Ex-2.16: The sum of all minterms of Boolean function of n variables is 1.

(a) prove the above statement for $n=3$.

Ex Solⁿ:

$$\begin{aligned} F &= a'b'c' + a'b'c + a'bc' + a'bc + ab'c' + ab'c \\ &\quad + abc' + abc \\ &= a'b' + a'b + ab' + ab \\ &= a' + a \\ &= 1 \end{aligned}$$

(b) suggest a procedure for a general proof:

Ex Solⁿ: Assume the statement is true for $n=k$, i.e. the sum of all minterms of a Boolean function of k variables is 1. we want to prove that the statement is also true for $n=k+1$ i.e. the sum of all minterms of a Boolean function of $k+1$ variables is 1.

To do so, we can use the fact,

that any Boolean function of $k+1$ variables can be expressed as a Boolean function of k variables using the following identity:

$$F(A_1, A_2, \dots, A_k, A_{k+1}) = F(A_1, A_2, \dots, A_k, 0)$$

$$+ (\text{not } A_{k+1}) + F(A_1, A_2, \dots, A_k, 1) \quad \text{--- } A_{k+1}$$

In other words, we can split the function into two parts: one that depends only on the first k variables and has $A_{k+1} = 1$. Each part can be expressed as a sum of minterms and the sum of all minterms of the original function is the sum of two parts.

↓

* Ex - 2.17: The product of all maxterms of a Boolean function of n variables is 0.

(a) prove the above statement for $n=3$.

Solⁿ: product of maxterms:

$$\begin{aligned} &= (a+b+c) (a+b+c') (a+b'+c) (a'+b+c) (a+b'+c') \\ &\quad (a'+b+c') (a'+b'+c) (a'+b'+c') \\ &= (a+b+c \cdot c') (a+b'+c \cdot c') (a'+b+c \cdot c') (a'+b'+c \cdot c') \\ &= (a+b) (a+b') (a'+b) (a'+b') \\ &= (a+b \cdot b') (a'+b \cdot b') \\ &= a \cdot a' \\ &= 0 \end{aligned}$$

(b) Suggest a procedure for a general proof. Can we use the duality principle after proving (b) of problem 2.16?

Ex 1 Soln: Let $f(x_1, x_2, \dots, x_n)$ be a Boolean function and let $m_1, m_2, \dots, m_k$ be its minterms. (18)

Then we have:

$$f(x_1, x_2, \dots, x_n) = m_1 + m_2 + \dots + m_k$$

The complement of $(x_1, x_2, \dots, x_n)$ is given by:

$$f'(x_1, x_2, \dots, x_n) = (x_1' + m_1') (x_2' + m_2') \dots (x_n' + m_k')$$

where, $m_1', m_2', \dots, m_k'$ are the complements of the minterms.

Now, consider the product of all maxterms:

$$m_1 \cdot m_2 \cdot \dots \cdot m_k$$

where m_i is a maxterm. We want to show that this product is equal to 0.

To do this, we can use De Morgan's laws to rewrite the complement of f as disjunction of minterms:

$$f'(x_1, x_2, \dots, x_n) = (m_1 \times m_2 \times \dots \times m_K)'$$

Expanding the complement of this expression using De Morgan's law and the complement of a product rule, we get:

$$f'(x_1, x_2, \dots, x_n) = m_1 + m_2 + \dots + m_K$$

Now, we can re-write the product of all maxterms as the products of the complements of all minterms:

$$M_1 \times M_2 \times \dots \times M_K = (m_1') \times (m_2') \times \dots \times (m_K')$$

Multiplying these expressions together, we get:

$$(m_1') \times (m_2') \times \dots \times (m_K') \times (m_1 + m_2 + \dots + m_K)$$

The reason this is true is because the product of any minterm and its complement

is 0.

⇒ we can use the duality principle to show that the statement about the sum of all minterms is equivalent to the statement about the product of all maxterms.

Specifically, we can apply the duality principle to the SOP form of the Boolean function used in the proof of the product of all maxterms to obtain the POS form, and then we can use the proof for the sum of all minterms to show that the product of all maxterms is 0. Therefore, the duality principle can be used to establish the equivalence between the two statements.

* Ex-2.18: Show that the dual of the Ex-OR is equal to its complement.

Soln:

$$A \oplus B = A'B + AB'$$

Dual of a function: Change the OR to AND and AND to OR operation.

$$\text{dual} = (A+B)(A+B')$$

$$\begin{aligned}\text{Compliment} &= (A'B + AB')' \\ &= (A'B)'(AB')'\end{aligned}$$

$$= (A+B')(A'+B)$$

which shows compliment and dual of Ex-OR are equal.

* Ex-2.19: By substituting the Boolean function equivalent of the binary operations as defined in Table 2-6,

Show that,

(a) The inhibition and implication operators are neither commutative and associative.

III Soln:

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$$\Rightarrow x/y = xy' \neq y/x = x'y \quad \text{Not commutative}$$

$$\begin{aligned} \Rightarrow (x/y)/z &= xy'z' \neq x/(y/z) = x(yz')' \\ &= x(y+z) \\ &= \text{Not Associative.} \end{aligned}$$

(b) The Ex-OR and equivalence operators are commutative and associative.

III Soln:

$$F = x \oplus y$$

$$= xy' + x'y = y \oplus x$$

$$= yx' + y'x$$

$$= \text{commutative}$$

$$F = (x \oplus y) \oplus z = \Sigma(1, 2, 4, 7)$$

$$= x \oplus (y \oplus z) = \text{Associative}$$

(c) The NAND operator is not associative.

III Soln: not associative means

$$(A \uparrow B) \uparrow C \neq A \uparrow (B \uparrow C)$$

$$L.S = (A \uparrow B) \uparrow C$$

$$= \overline{AB \cdot C}$$

$$= \overline{AB} + \overline{C}$$

$$= AB + \overline{C}$$

$$R.S = \overline{A \uparrow (B \uparrow C)}$$

$$= \overline{A \cdot (BC)}$$

$$= \overline{A} + \overline{BC}$$

$$= \overline{A} + B\overline{C}$$

$$L.S \neq R.S$$

Hence, NAND is not Associative.

(d) NOR and NAND operators are not distributive.

$$\xrightarrow{\text{NOR}} \Rightarrow \overline{A+B} + C \neq \overline{A+BC + B+AC}$$

$$\therefore L.S = 0, R.S = 1$$

$$\therefore L.S \neq R.S$$

Here,

$$A=1$$

$$B=1$$

$$C=1$$

NAND: Also NAND Gate is not distributive.

* E.x - 2.20: The majority gate is a digital circuit whose output is equal to 1 if the majority of the inputs are 1s. The output is 0 otherwise. By means of a truth table, find the boolean function implemented by a 3-input majority gate. Simplify the function.

Solⁿ: Truth table

A	B	C	output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Apply k-map for the given expression and we find the simplified

expression for the

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$$F = AB + AC + BC = \text{output}$$

* Ex-2.21: verify the truth table for the 3-input Ex-OR gate listed in fig. 2-8(c). List all eight combinations of x, y, z ; evaluate

$$A = x \oplus y; \text{ then evaluate } F = A \oplus z \\ = x \oplus y \oplus z$$

Solⁿ: The truth table for the 3-input Ex-OR gate is:

x	y	z	$A = x \oplus y$	$F = A \oplus z$
0	0	0	0	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	0	1

* E.X - 2.22 TTL SSI come mostly in 14-pin packages. Two pins are reserved for power supply and the other pins are used for input and output terminals. How many gates are enclosed in one such package if it contains the following types of gates.

(a.) 2-input EXOR gates.

Solⁿ: In 14-pin TTL SSI package two pins are used for power supply, which leaves 12 pins available for input and output terminals. The actual numbers of gates enclosed in a package may vary depending on the specific design.

Two 2-input Ex-OR gates can be enclosed in one package, which would use 8 pins. Therefore a 14-pin package can contain up to 4 such gates.

(b) 3-input AND gates:

Solⁿ: Three 3-Input AND gates can be enclosed in one package which would use 9 pins. Therefore a 14-pin package can contain up to 4 such gates.

(c) 4-input NAND gates:

Solⁿ: Two 4-input NAND gates can be enclosed in one package, which would use 8 pins. Therefore, a 14-pin package can contain up to 4 such gates.

(d) 5-input NOR gates:

Ans: One 5-input NOR gate can be enclosed in one package which would use 5 pins. Therefore, a 14-pin package can contain up to 2 such gates.

(e) 8-input NAND gates.

Ans: One 8-input NAND gate can be enclosed in one package, which would use 8 pins. Therefore, a 14-pin package can contain up to 1 such gate.

* E.X-2.23: Show that a positive logic AND gate is a negative logic OR gate and vice-versa

Solⁿ: A positive-logic AND gate can be represented by the following truth table:

A	B	output
0	0	0
0	1	0
1	0	0
1	1	1

⇒ truth table for this negative-logic OR gate will be:

A	B	output
1	1	1
1	0	0
0	1	0
0	0	0

Comparing this truth table to the original positive-logic AND gate truth table, we can see that they are identical except that the inputs and output have been inverted. Therefore, A positive logic AND gate is a negative-logic OR gate.

* E.X-2.24: An IC logic family has NAND gates with fan-out of 5 and buffer gates with fan-out of 10. Show ~~that~~ how the output signal of a single NAND gate can be applied to 50 other gate inputs.

Solⁿ: ~~Feed~~ To apply the output of a single NAND gate to 50 other gate inputs, we can use a combination of NAND ~~gate~~ and buffer gates in a tree-like structure.

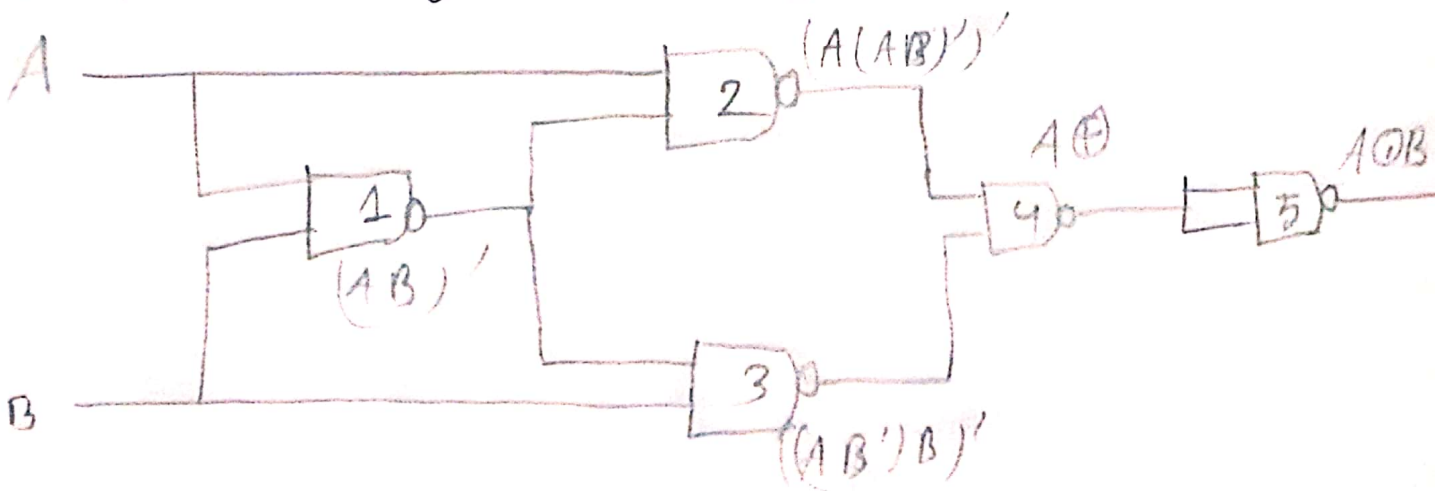
The output signal of the initial NAND gate is connected to the inputs of three NAND gates in parallel, each of which is followed by a buffer gate. The outputs

of these buffer gates are then connected to the inputs of three more NAND gates and so on, until the final level, where a single NAND gate is connected to 10 buffer gates, which drive the 50 output signals.

* E. x - 2.25: Implement Ex-NOR gate using minimum number of NAND gate and NAND gates using minimum number of NOR gates.

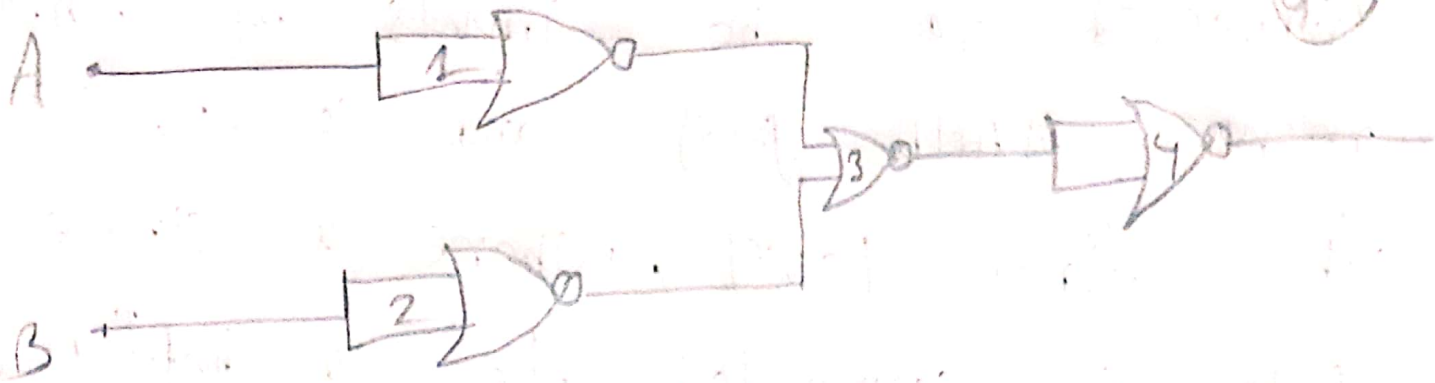
Solⁿ:

⇒ Ex-NOR gate using NAND gate:



⇒ NAND gate using NOR gates:

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* Ex - 2.26: Show that $(x \oplus y) \oplus z = x \oplus (y \oplus z)$.

Solⁿ: we can prove this using truth table:

x	y	z	$x \oplus y$	$x \oplus y \oplus z$	$y \oplus z$	$x \oplus (y \oplus z)$
0	0	0	0	1	1	1
0	0	1	0	0	0	0
0	1	0	1	0	0	0
0	1	1	1	1	0	1
1	0	0	1	0	0	0
1	0	1	1	1	0	1
1	1	0	0	1	0	1
1	1	1	0	0	1	0

$$\therefore x \oplus y \oplus z = x \oplus (y \oplus z)$$

Ans

* Ex-2.27 with examples, make out difference between. (93)

(a) min-terms and max term.

Solⁿ: $\Rightarrow$ min terms

min terms	max term
1. A method of describing a Boolean expression using a set of product terms.	1. A method of describing a Boolean expression using a set of sum terms.
2. we write the product terms for each input combinations that gives high (1) output	2. we write the sum terms for each input combination that gives low (0) output.
3. Final expression is obtained by adding the relevant product terms	3. Final expression is obtained by multiplying the relevant sum terms.

(b) positive and negative logic.

Solⁿ:

⇒ positive logic is defined as a high voltage level representing a logic 1 and a low voltage level representing a logic 0.

⇒ Negative logic is the reverse i.e. a low voltage level represents a logic 1 and a high voltage level represents a logic 0.

* Ex - 2.28: Differentiate between standard TTL, Schottky TTL, ECL and CMOS logic families in terms of fan-out, noise-margin, propagation delay and power dissipation.

Q. Soln:

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Specification	TTL	ECL	CMOS
Fan-out	10	25	>50
Noise margin	very good	Good	Excellent
propagation delay	1.5-33	1-4	1-200
power dissipation	1-22	4-55	1 @ 1MHz